



First and oldest

The very first and the oldest domain name to be registered was www.symbolics.com, on the 15th March, 1985.

Who came first?

Scientists finally concluded that the chicken came first, not the egg, because the protein which makes egg shells is only produced by hens.

NAS

HOW WE TESTED

Since these were NAS devices made for home use and SOHO environments it didn't make sense to set up an array of 20-40 clients in virtual machines bombarding the device simultaneously. The average usage scenario for these devices would involve 1-2 clients performing a high intensity data transfer process at any given time. So with this in mind we went with the following process.

Setup

Some of the NAS devices did come pre-loaded with the operating system installed on a set of hard drives. We updated the OS to the latest version and had a look at the features for tabulation. Each NAS device was interfaced with a router which was then connected to a client testing computer. The network links were ensured to be regular from the NAS device to the client computer. CAT 6 cables were used and the router which acted as the intermediary was also Gigabit capable.

For the performance test, only one volume was created on each NAS device. We went with RAID 0 to maximise performance. In a work environment, RAID 1 or RAID 5 would

be preferred for data protection but we weren't bothered with data protection in this scenario. After creating the volume we went on with disabling any features that would hamper performance, for example, most devices have the ability to spin down the hard drives when they are not in use. This would've drastically affected seek times and subsequently performance metrics.

NASPT

Intel's NAS Performance Toolkit is a well rounded benchmark for NAS devices since it makes use of traces that were generated from real world operations. These mimic conditions and operations that are most likely to have occurred over time with NAS devices. There are traces that emulate the following

- Copying files to the NAS
- Copying files from the NAS
- Copying directories to the NAS
- Copying directories from the NAS
- Content creation involving multiple content sources across multiple file types
- Office productivity simulated using applications

handling spreadsheets, documents, etc

- HD video playback simulated using single playback, 2x playback of the same video, 4x playback of the same video, playback and recording, single file recording etc.
- Photo album browser test which has multiple image files being accessed from a large and deep directory structure

These tests were run over and over again and the average transfer rates are recorded.

Backup Feature

Since each NAS has its own backup feature we plug in a USB 3.0 drive to test the data transfer rates during a backup operation.

Media Playback

We run multiple instances of high bit-rate video files and we look at the seek times and if the client has to buffer. This test is performed over a wired connection.

Finally all products were scored on the basis of all scores recorded across four different heads – performance, features, value for money and build quality.

3.0 ports, an eSATA port, the Gigabit Ethernet port and a Kensington lock slot. The interior drive cage is made of metal but the bay trays are plastic. The trays have clamps on the sides to prevent the hard drive from accidentally popping out when being serviced.

THECUS N2310/ N4310

The N2310 and the N4310 look quite similar from the front. The placement of all the components are the same but one crucial thing that is lacking is that there is no USB port on the front of the devices. Both have support for one-button backup and two USB 3.0 ports but the ports are on the rear and the button is on the front. This is a minor inconvenience for those who'd have to make use of this

particular feature regularly. The drive trays are spring loaded but there are no clamps to hold the drives onto the trays. There are small projections on the bottom of each tray which slot into the screw holes on hard drives and this prevents the drives from sliding on the tray. The N4310 has a 90mm fan while the N2310 has to make do with a 60mm fan. Both devices have vents on the front of the drive trays and the N4310 has one wide vent on the chassis itself.

The chassis of the N2310 is primarily plastic, even the Kensington slot is seated in the plastic body while the N4310 opts for metal on the sides and the rear. Both devices make use of power connectors which look similar to the ones used with laptops. We can see why Thecus has decided to vary the two devices so vastly,

the N4310 costs ₹32,456 which is a little over twice at which the N2310 is priced at.

NETGEAR RN104

NETGEAR's RN104 carries a price tag of ₹32,500 but you can get it from about ₹18,755 at multiple online retailers. It may be priced way lesser than the Synology but when it comes to build quality, the RN104 has a much

sturdier build. Aside from the front panel, everything else is made of metal. The rear even has a handle to grab onto and even without drives installed, this NAS unit is quite heavy so the handle is a nice touch. A 90mm fan on the rear is what drives cool air into the device which exits via vents that are situated towards the front of the device on the sides and the top. Being



Thecus N4310